

Chapter 18: Sound: Production of Sound

Textbook Exercise Solutions & Answers



1. Fill in the blanks:

(a) Sound is generated by the rhythmic of any object.

vibrations

(b) The frequency of sound is measured in

Hertz (Hz)

(c) If of sound is decreased its loudness also decreases.

intensity

(d) A medium is necessary for of sound.

generation

2. Match the pairs.

Column 'A' Column 'B' 1. Flute a. Frequency less than 20 Hz 2. Frequency b. Frequency more than 20,000 Hz 3. Sound level c. Vibrations in the air 4. Ultrasonic sound d. Measured in Hz 5. Infrasonic sound e. Decibel

Column 'A'	Column 'B'
1. Flute	c. Vibrations in the air
2. Frequency	d. Measured in Hz
3. Sound level	e. Decibel
4. Ultrasonic sound	b. Frequency more than 20,000 Hz
5. Infrasonic sound	a. Frequency less than 20 Hz

3. Give scientific reasons.

(a) In earlier times, people used to listen for the arrival of a distant train by putting their ear to the rails.

1. It was a technique that worked better in those days when train speed were slower than they are today and when there were more rail joints.
2. When people would put their ear to the rail they could hear vibration caused by the wheels moving over the joints. ,
3. Metal being a solid is an efficient medium for the sound to travel and so the sound can propagate further in the rail than would in air as speed of sound is faster in solids than in air.

(b) The sounds generated by a tabla and sitar are different.

1. In tabla where leather is fitted, sound gets generated when the leather surface is thumped by fingers or palm.
2. The waves thus created travel in two dimensions. Sitar is having strings attached to pumpkin box called sound box.
3. Sounds generated by tabla and a sitar are different because they have different pitch and frequencies.

(c) If you were both on the moon your friend will not be able to hear you call.

Yes, your friend will not be able to hear your call because sound needs a medium for transmission since it is a mechanical wave it cannot travel through vacuum.

1. As there is no air on the moon and sound is actually vibration in air.
2. So astronauts communicate to each other by signals or microphones in space even when they are closer.

(d) We can hear the movement of a mosquito wings but we cannot hear the movement of our hands.

1. Humans cannot hear frequency less than 20 Hz. In the movement of hands oscillations does take place but is less than frequency that humans can hear.
2. Mosquitoes wings flap in air.
3. When they fly in air it produces buzzing sound in our ears.
4. The movement of mosquito wings produces sound greater than 20 Hz.
5. Sound is the reception of such waves and their perception by the brain.

4. Write answers to the following questions.

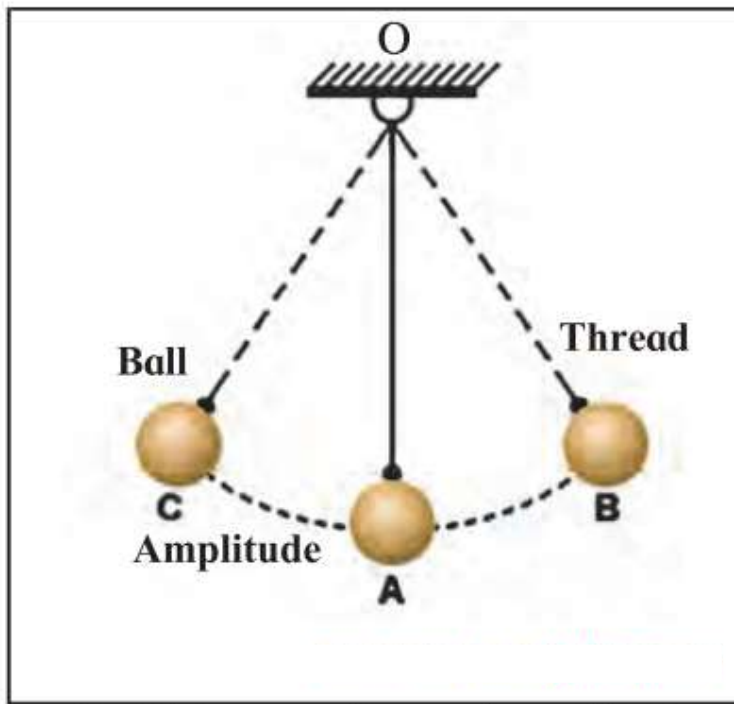
(a) How is sound produced?

1. Sound is generated due to the vibration of an object, e.g. bell, string or diaphragm of a musical instrument.
2. Vibrations are imparted to the molecules in the air. These vibrations reach our ears through the air and sound is heard.
3. Vibrations of an object is necessary for generation of sound.
4. These vibrations in air are called travelling longitudinal waves which we can hear.

(b) What does the intensity of sound depend upon?

1. The intensity of sound is proportional to the square of the amplitude of vibration, e.g. If the amplitude is doubled the intensity of sound becomes four times.
2. Therefore the loudness or softness of sound is dependent on the intensity of sound and sound level.

(c) Explain how the frequency of oscillation is related to the length of a pendulum and the amplitude of its oscillation.



Oscillatory motion and amplitude

1. Given an oscillatory motion to the pendulum, the maximum distance between the original position A of the pendulum and extreme position B or C is called the amplitude of oscillation. AB, AC is the amplitude of oscillation.
2. The number of oscillations completed by an oscillator in one second is called the frequency of oscillation.

Relation between frequency and length of a pendulum: If the length of the pendulum increases the period of oscillations also increases.

We know time period of oscillations = $\frac{1}{\text{Frequency}}$

If time period increases, frequency decreases, therefore, if the length of the pendulum increases frequency will decrease.

Relation between frequency and amplitude: Frequency of oscillation is independent of its amplitude of oscillation.

(d) Explain the two ways by which the pitch of the sound generated by a stretched string can be changed?

1. A string vibrates with a particular fundamental frequency. It is however possible to produce pitches with different frequencies from the same string.
 2. The two properties of the string that affect its frequency are length and diameter.
 3. When the length of a string is changed, it will vibrate with a different frequency. Shorter strings have higher frequency and therefore higher pitch.
 4. When a musician presses her finger on a string, she shortens its length. The more fingers she adds to the string, the shorter she makes it, higher will be the pitch.
- Diameter: Thick strings with large diameter vibrate slower and have lower frequency than the thin ones.